

UMNUGOVI, Mongolia

WMO: 442150

Lat: 49.1047N

Lon: 91.7261E

Elev: 1590

StdP: 83.63

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.0	-34.1	-39.3	0.1	-36.9	-36.3	0.1	-33.7	12.3	-11.2	9.8	-10.5	0.3	270	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	27.1	14.8	25.5	14.0	24.0	13.4	16.4	24.4	15.4	22.8	14.5	21.6	3.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	11.8	18.8	12.6	11.0	18.2	11.6	10.3	17.7	52.3	24.7	49.0	22.7	46.3	21.7	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	9.4	8.3	-36.9	29.9	4.5	1.6	-40.1	31.0	-42.7	32.0	-45.2	32.9	-48.5	34.0		
			-36.8	17.8	4.4	1.9	-40.0	19.1	-42.6	20.2	-45.1	21.3	-48.3	22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-22.3	-17.8	-6.8	3.2	9.7	16.0	17.9	15.5	8.8	0.7	-9.4	-18.0
	DBStd	14.74	6.49	7.38	6.69	5.18	4.52	3.36	2.80	3.30	4.35	4.94	6.13	6.25
	HDD10.0	4390	1003	779	521	210	61	3	0	3	72	288	583	868
	HDD18.3	6789	1261	1012	779	456	268	83	41	97	287	546	833	1126
	CDD10.0	693	0	0	0	5	52	182	245	174	35	1	0	0
	CDD18.3	51	0	0	0	0	1	12	28	10	0	0	0	0
	CDH23.3	457	0	0	0	1	12	119	223	96	6	0	0	0
	CDH26.7	64	0	0	0	0	1	13	37	12	1	0	0	0
Wind	WSAvg	2.4	1.2	1.5	2.7	3.5	3.5	3.0	2.6	2.3	2.5	2.5	2.1	1.6
Precipitation	PrecAvg	129	1	1	2	5	7	24	41	30	10	3	2	2
	PrecMax	250	5	3	7	19	32	69	108	66	39	20	13	13
	PrecMin	71	0	0	0	0	0	1	2	2	0	0	0	0
	PrecStd	42	1	1	2	6	8	17	24	17	10	5	3	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.8	1.1	12.2	19.2	25.0	28.4	30.1	28.4	23.9	15.9	8.9	0.6
		MCWB	-6.0	-3.6	4.2	9.2	12.0	14.5	16.0	15.2	12.3	7.3	3.5	-3.4
	2%	DB	-6.5	-1.8	9.0	16.5	22.2	26.2	27.6	25.9	21.2	13.3	5.3	-2.5
		MCWB	-8.7	-5.4	2.3	7.3	11.1	13.7	15.4	14.5	11.0	6.0	0.8	-5.6
	5%	DB	-9.4	-4.3	6.5	14.2	20.1	24.6	25.8	24.0	19.0	11.1	2.3	-5.4
		MCWB	-11.3	-7.3	0.7	5.9	10.0	13.3	14.5	13.8	10.0	4.5	-1.5	-7.9
	10%	DB	-12.4	-7.0	3.8	12.1	18.1	23.0	24.2	22.4	17.0	9.0	-0.4	-8.1
		MCWB	-13.7	-9.4	-0.9	4.7	9.0	12.6	13.9	13.1	9.1	3.1	-3.6	-10.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.0	-3.3	4.9	10.3	13.3	17.0	18.2	17.3	13.7	8.1	3.7	-3.2
		MCDB	-3.1	0.6	11.1	17.4	22.2	25.6	26.1	25.1	22.7	14.5	8.3	0.6
	2%	WB	-8.7	-5.3	2.6	7.8	12.0	15.4	16.8	15.7	11.8	6.5	0.9	-5.6
		MCDB	-6.5	-1.8	8.6	15.2	20.6	23.3	25.1	23.1	19.7	12.7	5.0	-2.6
	5%	WB	-11.2	-7.3	0.8	6.5	10.8	14.3	15.8	14.7	10.6	4.9	-1.4	-7.9
		MCDB	-9.4	-4.4	6.1	13.5	18.8	22.2	23.4	21.8	17.9	10.5	2.1	-5.5
	10%	WB	-13.8	-9.3	-0.9	5.1	9.6	13.4	14.9	13.9	9.4	3.3	-3.5	-10.0
		MCDB	-12.4	-7.1	3.7	11.7	17.1	21.1	21.9	20.7	16.2	8.6	-0.5	-8.1
Mean Daily Temperature Range	5% DB	MDBR	11.2	13.6	13.8	12.4	11.7	11.0	10.2	10.7	12.4	12.0	11.0	10.3
		MCDBR	12.8	14.8	15.0	14.6	14.5	13.5	12.9	13.1	15.1	14.7	12.7	11.5
		MCWBR	10.8	11.8	9.9	8.4	7.6	5.8	5.3	5.3	7.6	9.2	9.1	9.3
	5% WB	MCDBR	12.8	14.8	14.5	13.8	12.9	12.3	11.5	11.9	14.0	14.0	12.4	11.5
		MCWBR	10.9	11.9	9.8	8.5	7.4	6.1	5.6	5.6	7.5	9.2	9.0	9.4
Clear-Sky Solar Irradiance	taub	0.235	0.262	0.307	0.353	0.351	0.356	0.353	0.334	0.297	0.279	0.255	0.235	
	taud	2.342	2.285	2.225	2.204	2.265	2.335	2.373	2.402	2.455	2.429	2.359	2.315	
	Ebn at Noon	851	897	902	891	907	902	899	904	911	869	810	793	
	Edh at Noon	65	89	114	131	128	121	115	106	90	77	64	58	
All-Sky Solar Radiation	RadAvg	1.56	2.62	4.11	5.54	6.57	6.74	6.52	5.77	4.66	2.95	1.65	1.21	
	RadStd	0.08	0.11	0.20	0.20	0.38	0.29	0.36	0.23	0.18	0.17	0.13	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.05	N/A	-0.70	-1.31	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	-2.36	N/A	-0.90	-1.24	N/A	N/A	N/A	N/A

Nomenclature: See separate page